

UNIT I: NUMBER SYSTEMS, BOOLEAN ALGEBRA & GATE MINIMIZATION

Weightage: 25% | Syllabus Section 1

1. Radix Arithmetic & IEEE 754

Complements: r 's comp: $r^n - N$; $(r-1)$'s comp: $(r^n - 1) - N$. Subtraction: $M - N = M + (r's\ comp\ of\ N)$.

IEEE 754 (32-bit Single Precision):

- Sign bit S (1 bit)
 - Biased Exponent E (8 bits, Bias = 127)
 - Mantissa M (23 bits normalized $1.M$)
- Value: $V = (-1)^S \times (1 + M) \times 2^{E-127}$.

2. Boolean Theorems & De Morgan

Absorption Laws: $A + AB = A$; $A(A + B) = A$.

Consensus Theorem: $AB + \bar{A}C + BC = AB + \bar{A}C$.

Dual: $(A + B)(\bar{A} + C)(B + C) = (A + B)(\bar{A} + C)$.

De Morgan's Laws: $\overline{A + B} = \bar{A} \cdot \bar{B}$; $\overline{A \cdot B} = \bar{A} + \bar{B}$.

Universal Gates: NAND & NOR implement all boolean functions.

3. Karnaugh Map (K-Map) & Quine-McCluskey

Gray Coding in K-Map: Adjacent cells differ by exactly 1 bit (00, 01, 11, 10). Group sizes: $2^k \in \{1, 2, 4, 8, 16\}$.

Prime Implicants (PI): Group that cannot be combined with any other group. Essential Prime Implicant (EPI) covers at least one minterm not covered by any other PI.

Don't Cares (d): Treated as 1 to form larger octets/quads; otherwise ignored as 0.

UNIT II: COMBINATIONAL LOGIC CIRCUITS & MSI MODULES

Weightage: 25% | Syllabus Section 2

1. Adders, Subtractors & CLA

Half Adder: Sum $S = A \oplus B$; Carry $C = AB$.

Full Adder: $S = A \oplus B \oplus C_{in}$; $C_{out} = AB + C_{in}(A \oplus B)$.

Carry Look-Ahead Adder (CLA): Generates carries in parallel to eliminate ripple delay: Generate $G_i = A_i B_i$, Propagate $P_i = A_i \oplus B_i$. Carry equation: $C_{i+1} = G_i + P_i C_i$. Constant delay $O(1)$!

2. Multiplexers & Demultiplexers

MUX ($2^n \rightarrow 1$): n select lines route 1 of 2^n inputs.

Formula: $Y = \sum_{i=0}^{2^n-1} m_i I_i$. Universal function generator!

DEMUX ($1 \rightarrow 2^n$): 1 data input directed to 1 of 2^n

outputs based on select lines. Identical to Decoder with Enable!

Priority Encoder ($2^n \rightarrow n$): If multiple inputs are active, highest priority index is output. Valid bit V indicates activity.

3. Decoders, Encoders & PLDs

Decoder ($n \rightarrow 2^n$): Binary input activates exactly 1 of 2^n minterms (Active-High) or maxterms (Active-Low).

Programmable Logic Devices (PLDs):

- PROM: Fixed AND array, Programmable OR array.
- PAL: Programmable AND array, Fixed OR array.
- PLA: Programmable AND array, Programmable OR array (most flexible).

UNIT III: SEQUENTIAL LOGIC CIRCUITS, FLIP-FLOPS & REGISTERS

Weightage: 25% | Syllabus Section 3

1. Flip-Flop Characteristic & Excitation Table

Flip-Flop	Characteristic Equation	Excitation ($Q \rightarrow Q_{next}$: 00, 01, 10, 11)
SR Flip-Flop	$Q_{n+1} = S + \bar{R}Q_n$ ($SR = 0$)	$S : 0, 1, 0, X; R : X, 0, 1, 0$
JK Flip-Flop	$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n$	$J : 0, 1, X, X; K : X, X, 1, 0$
D Flip-Flop	$Q_{n+1} = D$ (Transparent Data)	$D : 0, 1, 0, 1$
T Flip-Flop	$Q_{n+1} = T \oplus Q_n$ (Toggle)	$T : 0, 1, 1, 0$

Race-Around Condition: Occurs in level-triggered JK when $J = K = 1$ and clock pulse width $t_p > t_{pd}$. Solved by: (1) Master-Slave JK, (2) Edge-triggering.

2. Registers & Synchronous Counters

Shift Registers: SISO, SIPO, PISO, PIPO. Universal Shift Register supports parallel load, shift right/left, hold.
Ring Counter: Straight circulation of single '1'. $MOD = n$.
Johnson (Twisted Ring) Counter: Inverted output fed back. $MOD = 2n$. Uses only n flip-flops!
Counter Design Pipeline: (1) State Diagram, (2) State Table, (3) Determine Flip-Flop Excitations via K-Map, (4) Logic Circuit Diagram.

UNIT IV: FINITE STATE MACHINES (FSM) & MEMORY ARCHITECTURE

Weightage: 25% | Syllabus Section 4

1. Mealy vs Moore FSM Models

Mealy Machine: Output depends on BOTH present state AND present inputs: $Y = \lambda(S, X)$. Glitches possible on input changes; generally requires fewer states.
Moore Machine: Output depends ONLY on present state: $Y = \lambda(S)$. Outputs are synchronized to clock; strictly glitch-free.
State Reduction: Equivalence partitioning / Implication table removes redundant equivalent states without altering I/O behavior.

2. Memory Hierarchy & Interfacing

SRAM vs DRAM: SRAM uses 6-transistor latch, ultra-fast, used in Cache. DRAM uses 1 transistor + 1 capacitor, dense & cheap, requires periodic refresh.
Memory Capacity Expansion: To construct $2^M \times W_2$ memory using $2^m \times W_1$ chips:
• Chips required: $N = \frac{2^M \times W_2}{2^m \times W_1}$.
• Address lines for decoding: $M - m$ connected to $(M - m) \rightarrow 2^{M-m}$ decoder.